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| **Fiscal Year:** | FY 2019 | **Task Last Updated:** | FY 04/14/2019 |
| **PI Name:** | Hyers, Robert Ph.D. | | |
| **Project Title:** | Unified Support for THERMOLAB - ISS, ICOPROSOL, and PARSEC (NNX16AB40G) | | |
| **Division Name:** | Physical Sciences | | |
| **Program/Discipline:** | | | |
| **Program/Discipline--Element/Subdiscipline:** | MATERIALS SCIENCE--Materials science | | |
| **Joint Agency Name:** | | **TechPort:** | No |
| **Human Research Program Elements:** | None | | |
| **Human Research Program Risks:** | None | | |
| **Space Biology Element:** | None | | |
| **Space Biology Cross-Element Discipline:** | None | | |
| **Space Biology Special Category:** | None | | |
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| **City:** | Worcester | **State:** | MA |
| **Zip Code:** | 01609-2280 | **Congressional District:** | 2 |
| **Comments:** | Formerly with University of Massachusetts at Amherst. | | |
| **Project Type:** | Flight | **Solicitation / Funding Source:** | 2009 ESA AO Biology on Sounding Rockets and Physical Sciences on All Platforms |
| **Start Date:** | 11/16/2015 | **End Date:** | 11/15/2020 |
| **No. of Post Docs:** | | **No. of PhD Degrees:** | 0 |
| **No. of PhD Candidates:** | 2 | **No. of Master' Degrees:** | |
| **No. of Master's Candidates:** | | **No. of Bachelor's Degrees:** | |
| **No. of Bachelor's Candidates:** | | **Monitoring Center:** | NASA MSFC |
| **Contact Monitor:** | Craven, Paul | **Contact Phone:** | 256-544-7649 |
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| **Flight Program:** | ISS | | |
| **Flight Assignment:** | | | |
| **Key Personnel Changes/Previous PI:** | | | |
| **COI Name (Institution):** | | | |
| **Grant/Contract No.:** | NNX16AB40G | | |
| **Performance Goal No.:** | | | |
| **Performance Goal Text:** | | | |
| | NOTE: Continuation of "Unified Support for THERMOLAB - ISS, ICOPROSOL, and PARSEC," grant NNX10AR71G with the same investigator, Dr. Robert Hyers.<br>Robert Hyers is U.S. Team Member on European Space Agencey (ESA)-sponsored projects, "Peritectic Alloy Rapid Solidification with Electromagnetic Convection (PARSEC)," "Thermophysical Properties of Liquid Metallic Alloys - Modelling of Industrial Solidification Processes and Development of Advanced Products (THERMOLAB - ISS)," and "Thermophysical properties and solidification behaviour of undercooled Ti-Zr-Ni liquids showing an icosahedral short-range order (ICOPROSOL)."<br><br>Funding is for Dr. Hyers's role as Team Member on the subprojects described below, associated with European Space | | |

**Task Description:**

Agency (ESA)-sponsored main projects as noted.

Main project: Peritectic Alloy Rapid Solidification with Electromagnetic Convection (PARSEC); Science Coordinator is Dr. Thomas Volkmann, German Aerospace Center DLR, Institut für Materialphysik im Weltraum, Linder Höhe, Köln, Germany

Subprojects:

1. Transformation Kinetics-ISS (International Space Station). Task description: Using results from thermophysical property evaluation and combining these with the Magnetohydrodynamic (MHD) convection definition, we will set convection over as wide a range as is possible and conduct recalescence studies from the undercooled melt using the Material Science Laboratory Electromagnetic Levitator (MSL-EML) to compare to results from ground-based EML and ESL. The critical parameters of interest are evaluation of the incubation time between formation of metastable phases and subsequent transformation to the stable phase for Fe-based and titanium aluminide alloys. Since this transformation is very rapid, the high-speed video system must be used to image the processes associated with nucleation and growth during phase selection.

2. Thermophysical Properties. Task description: The density, viscosity, and surface tension of the flight samples will be measured in flight for a range of temperatures. Density will be determined by the video metrology, and surface tension and viscosity by the oscillating drop method. This WP includes determination of parameter modifications required for these measurements as well as collection, reduction, and analysis of the resulting data. The resulting data will support the whole range of theoretical and modeling efforts in this proposal.

3. Modeling of Macroconvection. Task description: Magnetohydrodynamic (MHD) model of macroconvection in the samples for different combinations of experimental parameters. Evaluation of fluid velocity, shear rates, turbulence intensity, and their distribution for control of experiments and input to other theoretical and modeling efforts. Collaboration on development of parameters for other work packages to enable the use of convective state as an experimental parameter.

Main project: Thermophysical properties and solidification behaviour of undercooled Ti-Zr-Ni liquids showing an icosahedral short-range order (ICOPROSOL); Science Coordinator is Dr. Dirk Holland-Moritz, Deutsches Zentrum für Luft- und Raumfahrt (DLR), Institut für Materialphysik im Weltraum, Linder Höhe, Köln, Germany.

Subproject:

Modeling of macroconvection: Magnetohydrodynamic modelling of macroconvection in Ti-Zr-Ni samples for different combinations of experimental parameters. Evaluation of fluid velocity, shear rates, turbulence intensity, and their distribution for control of experiments and input to other theoretical and modeling efforts.

Main project: Thermophysical Properties of Liquid Metallic Alloys - Modelling of Industrial Solidification Processes and Development of Advanced Products (THERMOLAB - ISS); Science Coordinator is Professor Hans-Joerg Fecht, Ulm University, Ulm, Germany

Subprojects:

Long Duration Microgravity Experiments - ISS. Evaluation of microgravity experiments for fluid flow investigations.

MODELLING A: Thermodynamics, Phase Selection. Modelling of nucleation under various flow conditions for steel alloys.

MODELLING B: Non-contact thermophysical. Modeling of magnetohydrodynamic convection and shear in 1-g, microgravity for 4-6 alloys. Modeling of Marangoni convection in 1-g ESL drops for comparison with microgravity experiments. Modeling of temperature distribution in 1-g and microgravity samples, 4-6 alloys. Input into design of experimental parameters for 4-6 Alloys.

**Rationale for HRP Directed Research:**

**Research Impact/Earth Benefits:**

This project consists of modeling in support of three different flight experiments:
- PARSEC - Peritectic Alloy Rapid Solidification with EM Convection

- ThermoLAB - Thermophysical Properties of Liquid Metallic Alloys

- ICOPROSOL - Thermophysical properties and solidification behaviour of undercooled Ti-Zr-Ni liquids showing an icosahedral short-range order

In PARSEC, we are investigating the effect of fluid flow on the solidification path of certain alloys. Control of the solidification path would enable tailoring of the microstructure and properties of metal parts for specific applications.

In ThermoLAB, we are investigating the thermophysical properties of high-temperature materials. Many of these materials are used commercially. A better understanding of the physical properties will allow more efficient and more reliable production of metallic parts using these alloys. Other alloys are chosen because of scientific interest in their properties, which may illuminate fundamental physics related to the formation of bulk metallic glasses and quasicrystals.

In ICOPROSOL, we are investigating the nucleation and growth of quasicrystals, and the effect of atomic-scale order on the macroscopic properties of these alloys. This fundamental investigation may improve our ability to tailor the microstructure of metals for commercial applications.

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| **Task Progress:** | Highlights of Work Accomplished During Grant Period 2017-2018.<br>• Supported the magnetohydrodynamic calculations for the prediction of phase selection during solidification for PARSEC.<br><br>• Completed the magnetohydrodynamic calculations for analysis of the transition to turbulence in positioner-dominated flow in MSL-EML for SEMITHERM project (“Investigations of Thermophysical Properties of Liquid Semiconductors in the Melt and in the Undercooled State under Microgravity Conditions (IAO-2000-068)”).<br><br>• Submitted report to SEMITHERM team.<br><br>• Began preparation of a journal article on these results.<br><br>• Completed the magnetohydrodynamic calculations for analysis of the effect of flow on unusual growth of the solid in Al-Ni for NEQUISOL (“Non-Equilibrium Solidification, Dendrite Growth Velocities & Modelling for Microstructure Engineering of Industrial Alloys (AO-99-023/025/036)").<br><br>• Submitted report to NEQUISOL team.<br><br>• Submitted an abstract to Minerals, Metals & Materials Society (TMS) Annual Meeting to present these results. [Ed. note: abstract presented in March 2019]<br><br>• Began revised MHD calculations for QUASI and ICOPROSOL.<br><br>• Visited DLR Cologne from July 1 to August 15, 2018.<br><br>o Operated Batch 1.3 cycles on International Space Station (ISS).<br><br>o Performed synchrotron x-ray measurements on the structure of quasicrystal-forming alloys in the undercooled liquid state with ICOPROSOL at the European Synchrotron Radiation Facility.<br><br>o Collaboration with other DLR personnel on various aspects of the grant Unified Support for THERMOLAB - ISS, ICOPROSOL, and PARSEC (NNX16AB40G) (USTIP).<br><br>• Attended International Work Group Meeting on MSL-EML at DLR Bonn, June 12-14, 2018.<br><br>• Image processing code for the analysis of MSL-EML data is being adjusted (expect to process the data downloaded from ISS this fall). |
| **Bibliography Type:** | Description: (Last Updated: 03/30/2026) |
| **Articles in Peer-reviewed Journals** | Xiao X, Hyers RW, Wunderlich RD, Fecht H-J, Matson DM. "Deformation induced frequency shifts of oscillating droplets during molten metal surface tension measurement." Applied Physics Letters. 2018 Jul;113(1):011903. https://doi.org/10.1063/1.5039336 , Jul-2018 |
| **Articles in Peer-reviewed Journals** | Xiao X, Lee J, Hyers RW, Matson DM. "Numerical representations for flow velocity and shear rate inside electromagnetically levitated droplets in microgravity." NPJ Microgravity. 2019 Mar 25;5:7. eCollection 2019. https://doi.org/10.1038/s41526-019-0067-2 ; PubMed PMID: 30937361; PubMed Central PMCID: PMC6433928 , Mar-2019 |
| **Books/Book Chapters** | Bracker G, Xiao X, Lee J, Reinartz M, Burggraf S, Herlach D, Rettenmayr M, Matson DM, Hyers RW. "Modeling of Fluid Flow Effects on Experiments Using Electromagnetic Levitation in Reduced Gravity." in "Materials Processing Fundamentals 2019. (from the 2019 TMS Annual Meeting & Exhibition)" Ed. G. Lambotte, J. Lee, A. Allanore, S. Wagstaff. Cham: Springer International Publishing, 2019. https://doi.org/10.1007/978-3-030-05728-2_16 , Jan-2019 |
| **Books/Book Chapters** | Hyers RW, Zhao J, Bracker G, Wunderlich R, Fecht H. "Anomalous nucleation in undercooled melts processed by electromagnetic levitation." in "Light Metals 2019. (from the symposium at 2019 TMS Annual Meeting & Exhibition)" Ed. C. Chesonis. Cham: Springer International Publishing 2019. https://doi.org/10.1007/978-3-030-05864-7_198 , Jan-2019 |
| **Papers from Meeting Proceedings** | Wunderlich RD, Mohr M, Hecht U, Matson DM, Xiao X, Lee J, Hyers RW, Bracker G, Fecht H-J. "The Materials Science Laboratory - Electromagnetic Levitator (EML) on the International Space Station: Thermophysical properties of a TiAl alloy (GE 48-2-2) in the liquid phase." 69th International Astronautical Congress (IAC), Bremen, Germany, October 1-5, 2018.<br>In: Proceedings of the 69th International Astronautical Congress (IAC), Bremen, Germany, 1-5 October 2018. Paper IAC-18-F1.2.3; IAC-18.A2.6.13 , Oct-2018 |